

BELLE RIVER, ON, CanadaWMO: **716363**Lat: **42.300N**Lon: **82.700W**Elev: **602**StdP: **14.38**Time Zone: **-5.00 (NAE)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.7	10.1	-0.8	5.4	7.5	3.5	6.7	11.6	31.6	23.0	29.1	26.4	11.6	240	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	88.9	75.6	86.0	74.6	83.3	73.0	78.6	85.3	76.8	83.1	75.1	80.6	7.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	142.5	83.0	74.9	133.8	80.6	73.3	126.7	78.7	42.6	85.4	40.7	83.3	39.0	80.8	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.0	25.3	22.2	2.0	95.2	6.2	3.6	-2.4	97.8	-6.0	99.9	-9.5	102.0	-14.0	104.6		
(5)			1.4	82.2	6.0	2.4	-3.0	83.9	-6.5	85.3	-9.9	86.6	-14.3	88.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	25.9	29.8	36.8	48.9	59.1	69.6	73.9	71.6	65.8	53.6	42.9	32.0
	DBStd	18.43	10.32	8.48	9.82	8.43	7.86	7.50	5.04	4.79	6.91	7.63	8.08	9.55
	HDD50	2725	748	567	426	117	11	0	0	0	2	48	243	563
	HDD65	5952	1212	986	875	490	221	40	1	6	73	362	664	1022
	CDD50	3062	0	1	17	83	292	588	741	671	476	159	29	5
	CDD65	818	0	0	1	6	38	178	277	212	98	8	0	0
	CDH74	5132	0	0	4	54	263	1329	1885	1062	504	31	0	0
	CDH80	1431	0	0	0	6	59	454	571	243	96	2	0	0
Wind	WSAvg	9.5	10.9	10.3	11.3	11.3	8.8	7.3	7.5	7.9	8.4	9.0	10.5	10.4
Precipitation	PrecAvg	35.2	2.5	2.3	2.6	3.3	3.5	3.1	3.4	3.2	3.5	2.6	2.7	2.7
	PrecMax	55.6	4.1	5.3	5.6	5.5	6.7	5.3	6.5	6.6	7.3	5.7	7.0	5.4
	PrecMin	28.0	0.8	0.7	0.6	1.2	1.3	1.2	1.0	0.4	1.0	0.4	0.6	1.2
	PrecStd	5.6	0.9	1.1	1.1	1.1	1.7	1.3	1.4	1.5	1.9	1.2	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	60.6	72.5	80.4	86.1	92.2	93.5	89.9	87.0	77.6	65.2	61.3
		MCWB	54.8	56.8	61.6	67.3	70.4	77.3	79.1	76.7	72.4	67.9	58.6	57.3
	2%	DB	51.1	50.1	63.9	73.7	81.1	88.8	89.2	85.6	82.6	73.2	61.4	54.6
		MCWB	50.3	46.6	56.6	62.4	70.4	74.7	77.0	75.9	71.8	63.6	57.7	52.6
	5%	DB	44.5	44.5	57.4	67.2	76.3	85.7	85.8	82.5	79.7	69.2	58.4	48.3
		MCWB	42.6	42.1	52.6	58.6	65.8	73.8	75.3	73.8	70.2	61.9	55.5	46.0
	10%	DB	38.1	39.9	50.8	61.8	71.7	81.7	82.9	79.6	76.5	65.0	54.7	43.6
		MCWB	36.9	37.8	46.5	54.4	64.6	71.7	73.1	72.1	69.1	59.8	51.3	41.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	57.0	62.4	69.1	74.8	79.8	82.7	80.0	76.2	70.7	61.5	59.6
		MCDB	55.8	59.3	71.3	78.5	81.0	89.0	90.7	86.4	81.9	76.1	64.3	60.6
	2%	WB	50.7	47.0	57.4	63.9	71.5	77.4	79.1	77.5	74.3	66.3	58.7	52.5
		MCDB	51.0	49.3	62.2	70.3	77.9	84.7	85.3	83.7	79.6	70.7	60.9	54.5
	5%	WB	42.1	42.2	52.6	59.9	68.4	75.4	77.3	75.3	72.5	63.3	55.7	46.6
		MCDB	44.2	44.3	57.4	66.0	73.6	82.5	83.5	79.9	77.1	67.5	57.9	48.2
	10%	WB	36.9	38.0	46.5	55.3	65.3	73.5	75.4	73.6	70.3	60.6	52.1	41.8
		MCDB	38.0	39.6	50.2	59.9	70.4	80.0	80.9	77.9	74.5	64.4	54.1	43.3
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.4	13.1	14.8	15.7	15.5	14.9	13.2	14.4	12.8	10.4	9.3
		MCDBR	16.0	17.2	23.4	24.2	22.0	19.6	18.9	17.6	18.7	18.9	15.0	14.0
		MCWBR	14.9	14.4	17.0	15.9	12.8	9.3	8.6	9.4	9.5	10.9	12.7	12.0
	5% WB	MCDBR	15.8	16.9	22.7	21.8	19.9	17.7	17.5	15.7	16.2	16.0	13.6	13.8
		MCWBR	15.1	14.8	17.1	15.5	13.7	9.5	9.0	9.7	9.7	11.1	12.4	13.0
Clear-Sky Solar Irradiance	taub		0.303	0.332	0.353	0.397	0.430	0.445	0.435	0.425	0.386	0.359	0.320	0.311
	taud		2.398	2.292	2.336	2.249	2.207	2.209	2.262	2.279	2.363	2.421	2.492	2.429
	Ebn at Noon		266	273	282	276	269	264	266	265	268	262	259	252
	Edh at Noon		25	33	36	42	45	45	42	40	34	29	23	23
All-Sky Solar Radiation	RadAvg		447	727	1105	1470	1725	1947	1937	1685	1358	882	545	385
	RadStd		52	81	108	125	146	121	99	68	115	73	71	32